	Class	Adm No
Candidate Name: ______________________		

H

2015 Preliminary Examination 2
Pre-University 3

H2 Biology **9648/02**

Paper 2 Core Paper **17 September 2015**

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

The use of an approved scientific calculator is expected, where appropriate. You will lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

This question paper consists of 24 printed pages including 1 blank page.

[Turn over

For Examiner's Use

Section A

Answer **all** questions in this section.

1. The cell membrane is an important component for all cells. Its role is critical because its structural components provide the barrier that marks the boundaries of a cell. Fig. 1.1 shows part of the cell surface membrane of a liver cell.

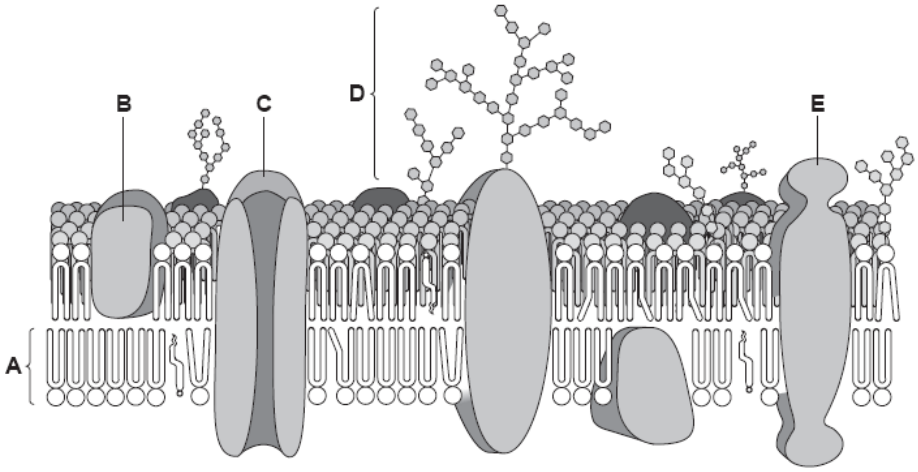

Fig. 1.1

(a) Identify biomolecules **A** and **C**.

A

A: Phospholipid;
C: Protein;
;; for 1 mark, max is 1 mark

C ……………………………………………..

[1]

(b) Describe the arrangement of biomolecules **A** in the membrane as seen in Fig. 1.1.

Phospholipids are arranged into lipid bilayer;
The hydrophilic phosphate heads of phospholipids face outwards into the aqueous environment inside and outside the cell + Interact with water molecules in the aqueous environment;
The hydrophobic hydrocarbon/fatty acid tails face inwards and create a hydrophobic interior + Repelled away from the aqueous environment;
; for 1 mark, max is 2 marks

………………………………………………………………………………………[2]

(c) Explain what determines the three-dimensional shape of biomolecule **C**.

- Specific sequence of amino acids made up the primary structure;
- Portions of the polypeptide chain form α- helix and β-pleated sheat (secondary structure);
- Secondary structures is formed by H bond between the C=O group and NH2 group of the polypeptide backbone;
- Tertiary structure is maintained by the hydrophobic interaction, ionic, hydrogen and disulphide bond formation between R-groups of amino acid residues;

; for 1 mark, max is 3 marks

For Examiner's Use

Biomolecule **D** is formed by joining many monosaccharides together.

(d) State the type of bond that joins monosaccharides together in biomolecule **D**

Glycosidic bond
; for 1 mark, max is 1 mark

[1]

A large number of storage polysaccharides can be found throughout the liver cell.

(e) Explain how the structure of this storage polysaccharide makes it a good energy storage molecule.

Glycogen is composed of several hundreds or thousands/many glucose molecules (linked by glycosidic bonds) +
Upon hydrolysis, the large number of glucose molecules, which are the main respiratory substrates/ are released to produce energy;
Most of the hydrophilic hydroxyl groups of the glucose residues project into the interior of the helices +
Hence glycogen is insoluble in water and can be stored in large quantities without affecting the osmotic potential of cells / do not easily diffuse out of cells;
Glycogen is highly branched +
Thus allowing it to be compacted, so that more glycogen can be stored in a given space / greater amount of glycogen can be stored per unit volume/ allows many enzymes to act on it at any one time, (hence can be easily hydrolysed to release glucose);
(correct structure linked to function)

; for 1 mark, max is 2 marks

[Total: 9]

For Examiner's Use

2. Gout is a medical condition, which is usually characterised by recurrent attacks of acute inflammatory arthritis. It is caused by the build-up of uric acid crystals in joints. Uric acid is produced from xanthine in a reaction catalysed by the enzyme xanthine oxidase.

A medical scientist investigated the effect of increasing xanthine concentration on the rate of activity of mammalian xanthine oxidase. Ten test-tubes were set up with each containing 5 cm^3 of different concentrations of xanthine solution. The test-tubes were placed in a water bath at 40 °C for ten minutes. A flask containing a xanthine oxidase solution was also put into the water bath. After ten minutes, 1 cm^3 of the xanthine oxidase solution was added to each test-tube. The reaction mixtures were kept at 40 °C for a further ten minutes. After ten minutes, the temperature of the water bath was raised to boiling point. The amount of uric acid was measured and used to calculate the rate of enzyme activity.

Results of the experiment are shown in Fig. 2.1.

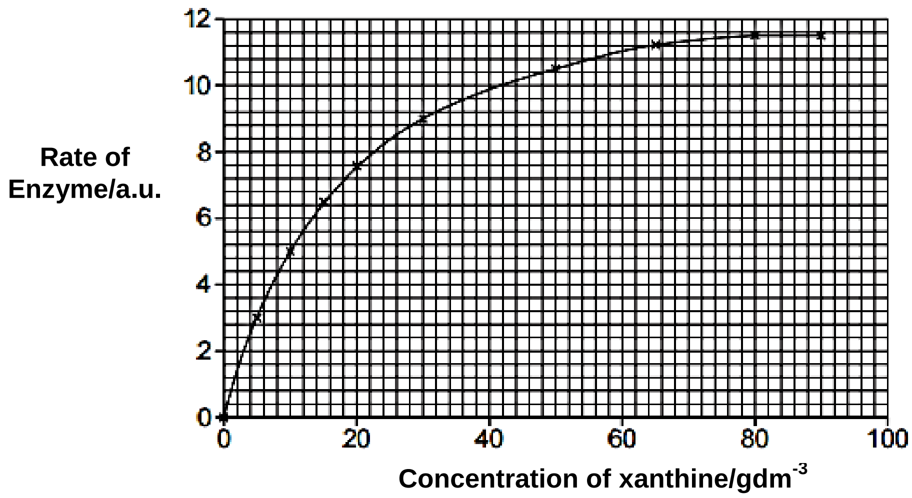

Fig. 2.1

(a) Account for the shape of the graph shown in Fig. 2.1.

<u>Description 1</u>
At low [xanthine], increase in [xanthine] results in an increase in the rate of enzyme reaction
; for 1 mark, max is 1 mark

<u>Explanation 1</u>
At low [xanthine], <u>not all the enzymes are bound to substrate molecules</u>/enzymes are not saturated, there are still <u>free active sites available</u> to bind to more substrate being added
[xanthine] is the limiting factor;
few collisions between enzyme and substrate, few, enzyme-substrate / E-S complexes formed ;
; for 1 mark, max is 1 mark

<u>Description 2</u>
<u>At high [xanthine] 80 - 90 gdm^{-3}</u>, rate of sucrose activity increases to a <u>maximum / plateau/ levels off/ remains constant at 11.4 – 11.6 au</u>;
; for 1 mark, max is 1 mark

<u>Explanation 2</u>
at high concentration / >80 g dm-3, [xanthine oxidase] is limiting factor;
maximum number of enzyme-substrate complexes formed ;
all the active sites are always occupied ;
enzymes are saturated;

For Examiner's Use

Using similar experimental procedures, the student proceeded on to investigate the effect of temperature on the activity of mammalian xanthine oxidase and xanthine oxidase found in thermophilic bacteria living in hot springs.

(b) Explain why xanthine oxidase in thermophilic bacteria has a higher optimum temperature than mammalian xanthine oxidase.

More disulfide bonds present in thermophilic sucrase compared to mammalian sucrase;
Ref to covalent nature and strength of disulfide bonds;
3D conformation of thermophilic sucrase is more stable than mammalian sucrase;
Thermophilic sucrase is more resistant to denaturation with increasing temperature;

; for 1 mark, max is 2 mark

Allopurinol is a drug used to treat gout. Fig. 2.2 shows the structures of xanthine and allopurinol.

Fig. 2.2

(c) With reference to Fig. 2.2, suggest how allopurinol can be used to treat gout.

Allopurinol has a similar shape/structurally similar/resemblance to xanthine
It enters active site / is a competitive inhibitor and prevents xanthine from binding to the enzyme active site;
Fewer ES complexes/fewer uric acid crystals formed/less uric acid formed;

; for 1 mark, max is 2 mark

....[2]

[Total: 8]

[Turn over

For Examiner's Use

3. Human T-cell Lymphotropic Virus (HTLV) was the first human retrovirus discovered. HTLV belongs to the Retroviridae family and predominantly affects T lymphocytes. HTLV has been implicated in several kinds of diseases including very aggressive adult T-cell lymphoma, which is the cancer of T-lymphocytes.

A molecule of messenger RNA (mRNA) was produced during the transcription of a HTLV gene in the host T lymphocytes. Part of the template sequence of DNA was ATGC. Fig. 3.1 shows the part of the molecule of messenger RNA corresponding to that sequence of four bases.

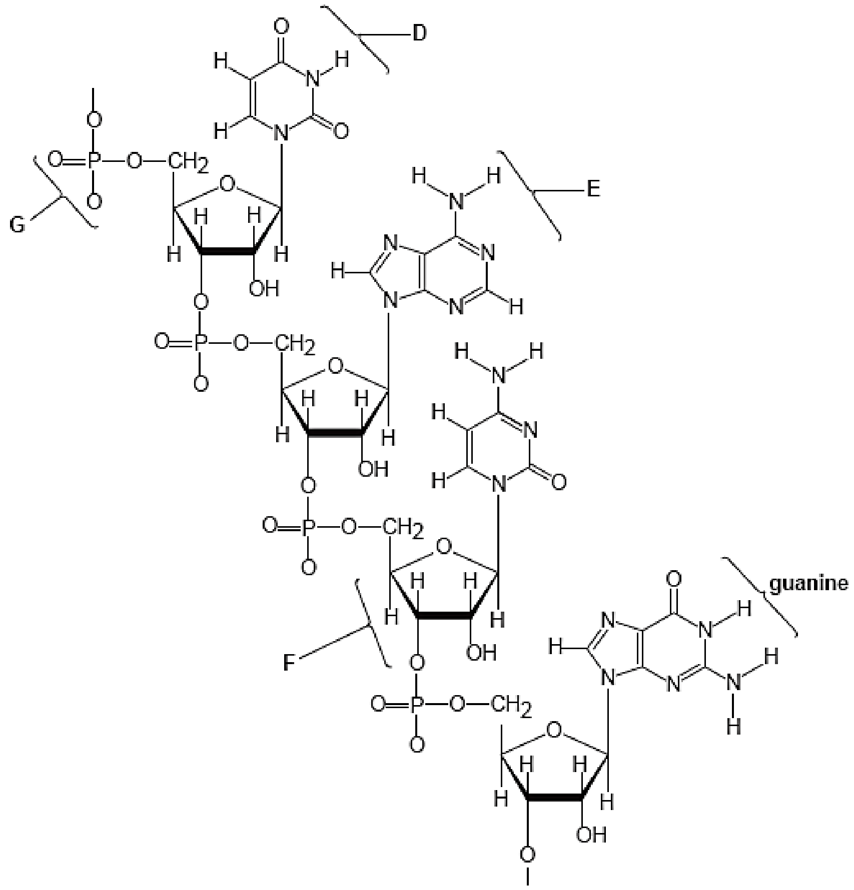

Fig. 3.1

(a) Name the parts of the mRNA molecule shown in Fig. 3.1 labelled **D**, **E**, **F** and **G**.

D: ......

E: ......

F: ......

G: ...... [2]

D – uracil ;
E – adenine ;
F – ribose ;
G – phosphate ;

;; for 1 mark, max is 2 marks

(b) Describe the role of mRNA after it leaves the host cell nucleus and enters the cytoplasm.

Provide a template for translation of viral proteins;
use of nucleotide / base, sequence, to make viral amino acid chain / polypeptides + reference to capsid proteins/viral enzymes;
mRNA binds/combines with ribosome + ref to small and/or large ribosomal sub-units;
Transfer / tRNA, bringing specific amino acid(s) to mRNA / ribosome;
Codon in the mRNA binds to anticodon on the tRNA by complementary base pairing + any e.g. of codon:anticodon base pairing ;
ref to polyribosome(s) / used by many ribosomes ;
mRNA short-lived + ref to production of viral protein for short period of time;

for 1 mark, max is 3 marks

…………………………………………………………………………………………. [3]

Fig. 3.2 shows the reproductive cycle of HTLV, which closely resembles that of HIV. HTLV binds to the GLUT-1 receptors on the T lymphocytes. Its viral envelope contains two types viral glycoproteins (SU and TM) deriving from a common precursor.

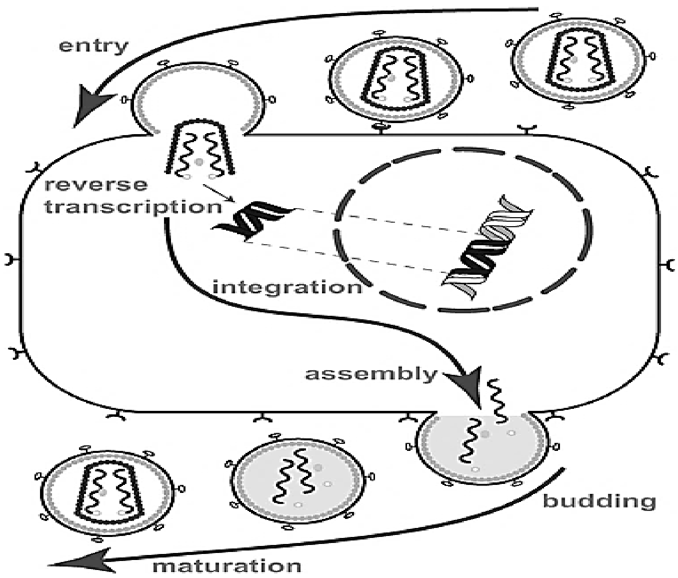

Fig. 3.2

(c) With reference to Fig. 3.2, distinguish between the mode of entry of a T4

Lytic phage	HTLV-I
Tail fibre of T4 binds to complementary specific receptors on bacterial cell membrane. (R: Base plate pin)	Glycoprotein (SU & TM) binds to GLUT-1 receptor on T lymphocytes cell surface membrane. Accept when there is mention of complementary receptors at least once.
Contraction of tail sheath/ The tail core protein is driven through the wall and the membrane + phage DNA injected.	Fusion of the viral's envelope (reject: membrane) with the host's cell membrane and capsid enters. Many thought that the process of entering the cell is receptor mediated endocytosis .

for 1 mark, max is 2 marks

For Examiner's Use

HIV spreads through body fluids and affects CD4$^+$ T lymphocytes. Over time, HIV can destroy so many of these cells that the body cannot fight off infections and diseases.

(d) Describe the importance of reverse transcriptase in HIV reproduction.

Virus uses reverse transcriptase to transcribe the RNA template into a complementary single stranded DNA;
Reverse transcriptase (Not: DNA polymerase) then synthesises a complementary DNA strand, forming a double-stranded DNA./allow for complementary base pairing to form a double-stranded DNA.
So can be integrated into the host double stranded DNA/genome/chromosome as provirus;
Reference to latency stage;

for 1 mark, max is 2 marks

Various drugs have been developed to treat HIV infections. Table 3.1 gives information about some of these drugs. HIV-1 surface glycoproteins are known to display antigenic variation due to antigenic drift. As a result, people, who receive drug treatment for HIV, have to take a mixture of drugs that act in different ways.

drug	**enzyme inhibited**
zidovudine	reverse transcriptase
tenofovir	reverse transcriptase
efavirenz	reverse transcriptase
atazanavir	protease

Table 3.1

(e) Using your knowledge of antigenic drift, explain the advantage of taking a mixture of the drugs shown in Table 3.1.

HIV has a high mutation rate;
Mutation in the viral genes coding for the viral enzymes alters the active site/ allosteric site/ tertiary structure of enzyme so drug cannot binds;
Virus may be resistant to one or more of the drugs / very low chances that HIV is resistant to all of the drugs;

for 1 mark, max is 2 marks

For Examiner's Use

Adult T-cell leukaemia/lymphoma (ATL) is usually a highly aggressive non-Hodgkin's lymphoma. Several lines of evidence suggest that HTLV causes ATL. This evidence includes the frequent isolation of HTLV-1 from patients with this disease and the detection of HTLV-1 proviral genome in ATL leukemic cells.

(f) Explain how the integration of the HTLV genome into T-lymphocytes' chromosome could lead to cancer.

Integration of viral DNA into the host chromosome**)** disrupts the proto-oncogene, converting it to oncogene + gain of function mutation;
Hyperactive proteins produced leading to excessive cell division/ excess amount of normal proteins to be produced leading to excessive cell division (if promter/enhancer affected)
or
integration disrupts/insertional mutagensis a t tumour suppressor gene + LOF mutation
no longer encode for the protein to inhibit cell division/ for apoptosis → uncontrolled cell division.

for 1 mark, max is 2 marks

...

...

...

[2]

The median age at diagnosis for ATL is 56 years.

(g) Using your knowledge on cancer development, suggest why ATL rarely appears in younger individuals.

More than one tumour suppressor gene/proto-oncogene needs to be mutated to lead to cancer development;
Time for accumulation of mutations in tumour suppressor genes and proto-oncogenes;
Time for reactivation of telomerase gene, tumour angiogenesis, invasion and metastasis;

for 1 mark, max is 1 mark

...

[1]

[Total: 14]

4. Penicillin is the first antibiotic to be discovered in 1928 by Alexander Fleming. Since then, antibiotics are being manufactured at an estimated scale of about 100,000 tonnes annually worldwide, and their use had a profound impact on the life of bacteria on earth. More strains of pathogens have become antibiotic resistant.

Penicillin resistance in *Streptococcus pneumoniae* involves point mutations in the gene coding for penicillin-binding proteins (PBPs). PBPs are the target enzymes of penicillin action. These mutations could result in reduced penicillin affinity.

(a) Describe how a mutation in *PBP* gene may result in an enzyme with reduced penicillin activity.

Missense mutation + Different codon/mRNA sequence encoded for different amino acid encoded for / different primary structure of PBP polypeptide;
Different R group (as original amino acid)/different forces of interactions (give at least 2 examples);
Different 3D configuration of PBP polypeptide /enzyme + Different 3D shape of active site;
Different affinity of active site for penicillin;

for 1 mark, max is 2 marks

[2]

Some strains of bacteria have become resistant to practically all of the commonly available antibiotics. A notorious case is the methicillin-resistant *Staphylococcus aureus* (MRSA). It can act as a major source of hospital-acquired infections. An old antibiotic, vancomycin, was resurrected for treatment of MRSA infections. Molecular studies revealed that a plasmid-borne transposable element encodes enzymes and regulator proteins that confer resistance of MRSA to the antibiotic vancomycin.

(b) Describe two ways by which vancomycin resistance may spread naturally amongst bacteria.

<u>Transformation</u>
Release of plasmid (containing vancomycin resistance gene) into environment;
Uptake of plasmid by bacteria cells;
<u>Conjugation</u>
Contact between recipient F^- and donor F^+ cells + involve formation of sex pilus;
Transfer of vancomycin resistance gene found on F plasmid;
<u>Transduction</u>
Vancomycin resistance gene from the host/donor bacterium is transferred to the recipient bacterium cell via bacteriophage;
Donor DNA from the host is accidentally packaged within viral capsid + progeny phages infect other bacteria;

for 1 mark, max is 2 marks

In the absence of glucose, *Escherichia coli* can proliferate using the pentose sugar arabinose. The ability of *E. coli* to utilise the sugar arabinose is regulated via the arabinose operon, depicted in Fig. 4.1 below.

The structural genes (*araA*, *araB*, and *araD*) encode enzymes for the metabolism of arabinose. The *araC* gene encodes a gene regulatory protein that binds adjacent to the promoter of the arabinose operon.

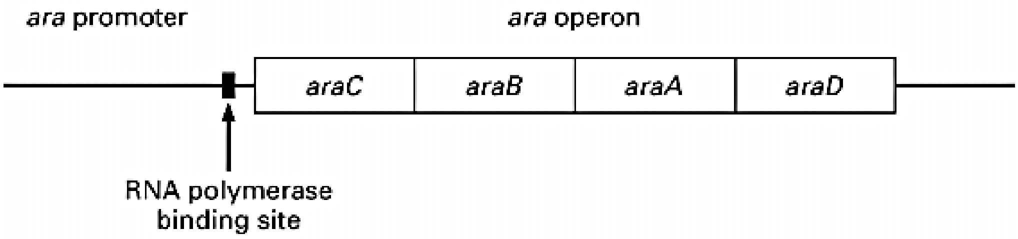

Fig. 4.1

(c) Distinguish between structural and regulatory genes.

....

A structural gene is a region of DNA that codes for a protein that forms part of a structure or has an enzymatic function;
A regulatory gene codes for a specific protein product that regulates the expression of the structural genes;

for 1 mark, max is 2 marks

For Examiner's Use

An experiment was conducted to investigate the role of araC regulatory protein in regulating arabinose operon. The results of the experiment are shown in Table 4.1.

Genotype	*araA* RNA Levels	
	in the absence of arabinose	in the presence of arabinose
araC+ (normal cells)	low	high
araC− (mutant cells)	low	low

Table 4.1

(d) With reference to the results in Table 4.1, explain whether araC regulatory protein functions as a repressor or activator in regulating the arabinose operon.

araC regulatory protein serves as a activator;
In the absence of functional araC protein, low/ no transcription/expression of ara A gene was observed;
araC protein when bound to arabinose activate the transcription of the structural genes;

for 1 mark, max is 2 marks

……………………………………………………………………………………[2]

[Total: 8]

For Examiner's Use

5. The human β globin locus is located on chromosome 11. This chromosome contains not only the β globin gene but also ε, γ_G, γ_A and δ globin genes. Expression of all of these genes is controlled by single locus control region (LCR), and they are differentially expressed throughout human development. β-LCR is a distal control element that upregulates the expression of these structural genes. Fig. 5.1 shows the various globin genes located on chromosome 11.

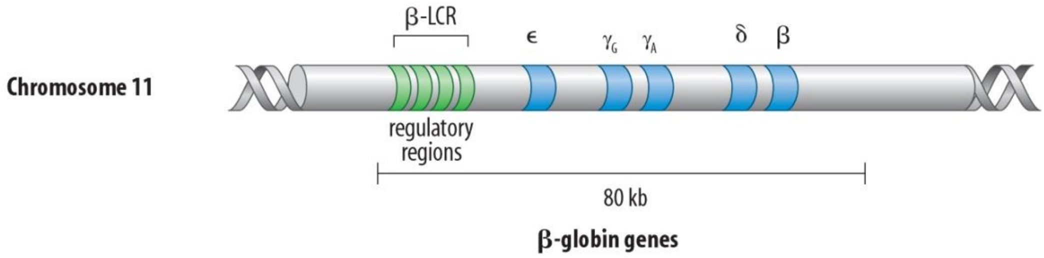

Fig. 5.1

(a) Explain how β-LCR regulates the expression of structural genes.

β-LCR is an enhancer sequence where activator proteins bind to increase / maximise the transcription of β gene;
Upon binding by the activator, DNA bending proteins facilitates DNA bending to bring activator closer to the promoter sequence;
ref. to recruitment of RNA polymerase and General Transcription Factors to form the transcription initiation complex;
Increases stability of TIC + affinity of RNA polymerase binding to the promoter;

for 1 mark, max is 3 marks

……………………………………………………………………………………………[3]

δ globin gene is expressed at a low level in adults. The low expression level of the gene is regulated by chromatin modification.

(b) Describe one way by which chromatin modification can lead to low level of δ-

DNA methylation is the covalent modification by addition of methyl groups to the promoter of a gene / cytosine nucleotides/ CpG dinucleotides after DNA replication;
catalysed by DNA methyltransferases;
changes the 3D conformation of DNA and prevents the binding of transcription factors to the promoter which is required for transcription initiation;
or
Deacetylation of acetylated lysine residues in histone tails + catalysed by histone deacetylases;
lysine residues become positively charged results in increased ionic interaction between histone and the linker DNA;
the chromatin structure becomes more compact / prevents access of transcription factors to the promoter which is required for transcription initiation;

for 1 mark, max is 2 marks

For Examiner's Use

The expression of globin genes in human can also be regulated at translational and post-translational level.

(c) Describe one way by which gene expression can be regulated at translational level.

Presence of translation initiation factors;
Facilitates the initiation of translation;
OR
Binding of translational repressors/ regulatory proteins to 5'UTR/ covers AUG;
Prevents the initiation of translation/ assembly of ribosome;
OR
Polyadenylation/ Addition of poly(A) tail increase stability of mRNA
A longer polyA tail increases the half -life of mRNA so more translation can occur;

for 1 mark, max is 2 marks

Haemoglobin and collagen are two of the most abundant proteins present in human.

(d) Distinguish between the molecular structures of haemoglobin and collagen.

Collagen	**Haemoglobin**
Molecule is <u>fibrous</u> in shape.	Molecule is globular in shape.
The primary structure consists of about 1000 amino acids residues	141 amino acids (α-globin chain subunit) and 146 amino acids (β-globin chain subunits)
Tertiary structure is not present/ secondary structure is most prominent	Primary, secondary, tertiary and, quaternary structures are present
Quaternary structure of tropocollagen consists of 3 polypeptide chains	Quaternary structure of haemoglobin consists of 4 polypeptide chains held together
Polypeptide chain cross-linked by hydrogen bonds to form a triple helix(tropocollagen)/ covalent bonds between tropocollagen to form fibrils	by hydrophobic interactions, ionic and hydrogen bonds.
Absence of prosthetic groups	Prosthetic group (haem group) associated with the protein
Sequence regularity (glycine is the 3rd amino acid)	Absence of sequence regularity

; for 1 mark, max is 2 mark

6. Summer squash are a subset of squashes that are harvested when immature while the rind is still tender and edible. Nearly all summer squashes are varieties of *Cucurbita pepo*. The colour of squash fruits may be white, yellow or green. The colours are determined by two different genes located on separate chromosomes. Coloured pigments are synthesised enzymatically from a white precursor as shown in Fig. 6.1.

White Precursor → Intermediate Compound → Green/Yellow compound

Enzyme 1 **Enzyme 2**

Fig. 6.1

When pure-bred white plants cross with pure-bred green plants, the resulting F_1 offspring are all white. When the resulting F_1 offspring undergoes self-fertilisation, the results of F_2 generation are as follow:

White plants: 183
Green plants: 42
Yellow plants: 15

(a) Construct a genetic diagram to explain the self-cross of F_1 generation and the results for the F_2 generation.

W represents the dominant allele at the first locus that code for inhibitor that inhibit the enzyme that produces the intermediate.
w represents the recessive allele at the first locus that do not code for the inhibitor
G represents the dominant allele at the 2^{nd} locus that codes for the enzyme that produces the green pigment
G represents the recessive allele at the 2^{nd} locus that codes for the enzymes that produces the yellow pigment

Parental phenotype	**White flower**	**X**	**White flower;**
Parental genotypes	**WwGg**		**WwGg**
Gametes;	**WG Wg wG wg**		**WG Wg wG wg ;**

Gametes	**WG**	**Wg**	**wG**	**wg**
WG	**WWGG White**	**WWGg White**	**WwGG White**	**WwGg White**
Wg	**WWGg White**	**WWgg White**	**WwGg White**	**Wwgg White**
wG	**WwGG White**	**WwGg White**	**wwGG Green**	**wwGg Green**

For Examiner's Use

wg	**WwGg** **White**	**Wwgg** **White**	**wwGg** **Green**	**wwgg** **yellow**

;

F_2 Genotypes:	WWGG, WWGg, WWgg, WwGG, WwGg, Wwgg,	wwGG, wwGg,	wwgg,	; correct genotype and phenotype
F2 Phenotype:	White	Green	Yellow	

F_1 Phenotypic ratio: 12 white: 3 green: 1 yellow;

[4]

(b) State the relationship between the two gene loci.

Dominant Epistasis;

; for 1 mark, max is 1 mark

..............................[1]

(c) Explain how you could determine that a white coloured plant is heterozygous at both loci.

The white-coloured plant had to be crossed with a yellow-flowered plant, which is homozygous recessive;
With reference test cross;
if it was heterozygous at both loci, the cross would give progenies that were white, green and yellow coloured;

; for 1 mark, max is 1 mark

In an investigation, the self-fertilisation of a white coloured plant and the results of the F_2 generation were analysed. A chi-squared analysis was carried out and the chi-squared value calculated was 2.10.

D.O.F	Probability, p				
	0.1	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 6.1

(d) Using the values in Table 6.1, draw appropriate conclusions as to whether the results of the cross actually followed the expected ratio you have predicted in **(a)**.

At $p = 0.05$ and degrees of freedom = 2, critical $\chi 2$ = 5.99;
Calculated $\chi 2$ is 2.1 and is less than critical $\chi 2$ value ;
Therefore the deviation between the observed and expected ratios is statistically insignificant + the difference is due to chance alone ;
Hence, H^o is not rejected and the observed results follows the 12:3:1 ratio ;

; for 1 mark, max is 3 marks

…………………………………………………………………………………………

…………………………………………………………………………………………[3]

For Examiner's Use

Fig. 6.2 shows the electron micrograph of a *C. pepo* cell undergoing a particular stage of meiosis.

Fig. 6.2

(e)

(i) Describe one event that occurs in Fig. 6.2.

Centromeres divide and each sister chromatid becomes a chromosome (R:split) ;
Spindle fibre attached to the centromere at the kinetochore protein ;
Spindle fibre shorten/Pulled by spindle fibre to the opposite poles of the cells (R:ends);
; for 1 mark, max is 1 mark

(ii) Explain how the second division of meiosis differs from mitosis.

In 2^{nd} division of meiosis, two sister chromatids of each chromosome are not genetically identical to each other vs genetically identical sister chromatids in mitosis;
Due to crossing over that occurs between homologous chromosomes during prophase I of meiosis. No crossing over occurs during prophase of mitosis.

; for 1 mark, max is 2 marks

Table 6.2 shows the number of chromosomes and the mass of DNA in different nuclei. All nuclei come from the same cell.

(f) Complete Table 6.2.

Nucleus	Number of chromosomes	Mass of DNA / arbitrary units
At prophase of mitosis	40	60
At telophase of mitosis	**40;**	**30;**
From a sperm cell	**20;**	**15;**

;; for 1 mark, max is 2 marks

[Total: 15]

For Examiner's Use

7. Bats represent the only known example of flight in mammals. They are the only animals with sophisticated laryngeal echolocation that is where ultrasonic calls are produced in the larynx. Many species of bats hunt flying insects at night. They are able to use echolocation (sound waves) to help them find their prey in the dark.

The oldest known fossilised remains of bats (50-60 million years ago) show that ancestral bats looked very much like they do today. However, fossil evidence is inconclusive whether or not the oldest species known to date (*Onychonycteris finneyi*) was capable of echolocation.

(a) Explain how the ability to use laryngeal echolocation may have evolved from an ancestral species that cannot use echolocation.

Pre-existing variations in the ancestral bat population due to random/spontaneous mutation;
Selection pressure is the competition for food/prey;
Individuals with the allele for echolocation have a selective advantage and selected for;
Reproduce and pass on the favourable alleles to the next generation;
Over many generations, increased frequency in the allele for echolocation;
Directional selection;

; for 1 mark, max is 3 marks

[3]

The geographical distribution of *Pipistrelles Pipistrellus* (common pipistrelle) extends across most of Europe, North Africa, and south-western Asia. It is one of the most common bat species in the British Isles. Originally, it was thought that all pipistrelle belong to the same species, *Pipistrelles Pipistrellus.* In 1999, the common pipistrelle was split into two species on the basis of different-frequency echolocation calls. The common pipistrelle uses a call of 45 kHz, while the *Pipistrellus Pygmaeus* (soprano pipistrelle) echolocate at 55 kHz.

Since the two species were distinguished, a number of other differences, in appearance, habitat and food, have also been discovered. It has been postulated that the distinct phenotypic differences between *Pipistrellus Pipistrellus* and *Pipistrellus Pygmaeus* may have arisen in the absence of geographical barriers.

For Examiner's Use

(b) Explain why molecular homology is better than anatomical homology in determining evolutionary relationship between the two species.

Similarity in anatomical features could be due to convergent evolution;
Study of anatomical homology is not possible between morphologically different species;
Using molecular method, organisms can be compared even if they are morphologically very different + all organisms have certain molecular traits in common, e.g. rRNA sequences or certain fundamental proteins;
Using morphological features as a benchmark is subjective and non-quantifiable;
Molecular data is quantifiable and objective. Nucleic acid and amino acid sequence data are precise and easy to quantify, hence allows an objective assessment of evolutionary relationships;
Fossils obtained from ancestral species may be incomplete thus comparative study of anatomy is not possible;
Compare molecular divergence of ancestral species with incomplete/no fossil record with that found in other lineages with more complete fossil records;

; for 1 mark, max is 2 marks

geographical barriers.

Sympatric speciation could have occurred through gene flow disruption between populations of original pipistrelles species;
Certain echolocation call/mating call of male pipistrelle are also selected for by female members;
Reference to non-random mating;
Differing conditions existing at different parts of Europe exerts different selection pressure. E.g. Availability of food at different habitats, different climates, creates different ecological niches;
Population of the original species exploiting a new niche may automatically reduce gene flow with other populations exploiting other niches;
Reference to habitat isolation;

; for 1 mark, max is 2 marks

Differences in the *cytochrome b* DNA sequence of several bat species from different regions of Europe were measured and plotted against time since divergence from the primitive ancestor (MYA) as seen in Fig. 7.1.

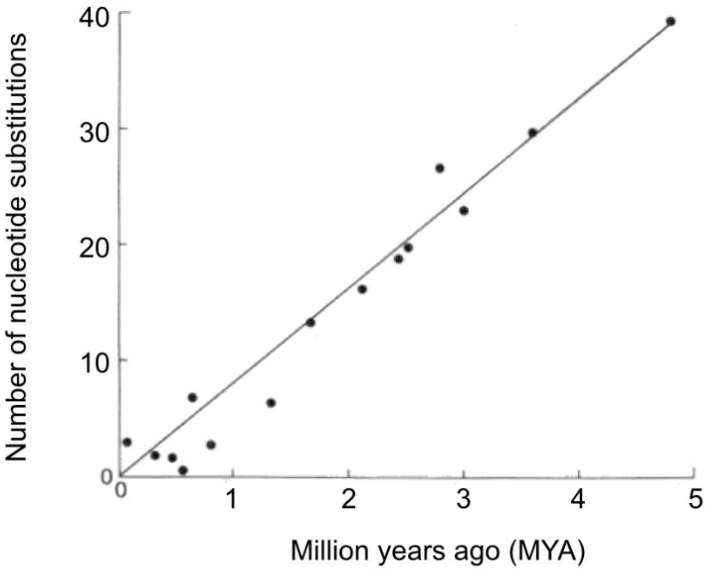

Fig. 7.1

(d) Describe how these differences support the neutral theory of molecular evolution.

Changes in the nucleotide sequence arise through neutral mutation;
for e.g. silent mutation or missense mutation where the change in amino acid does not occur in a critical region of the enzyme;
There is no effect on the phenotype and fitness of organism and thus allowed to accumulate;
Small number of changes over millions of years indicating that rate of mutation is slow as cytochrome b gene is a crucial gene in living organisms;
The plot of the line is straight, indicating that the rate of mutation is constant;

; for 1 mark, max is 3 marks

……………………………………………………………………………………………

……………………………………………………………………………………………[3]

[Total: 10]

For Examiner's Use

8. Fig. 8.1 shows a post-synaptic neurone, **Z**, forming synapses with adjacent neurones including Neuron **A** and **B**. The arrow in Fig. 8.1 shows the direction of the nerve impulse.

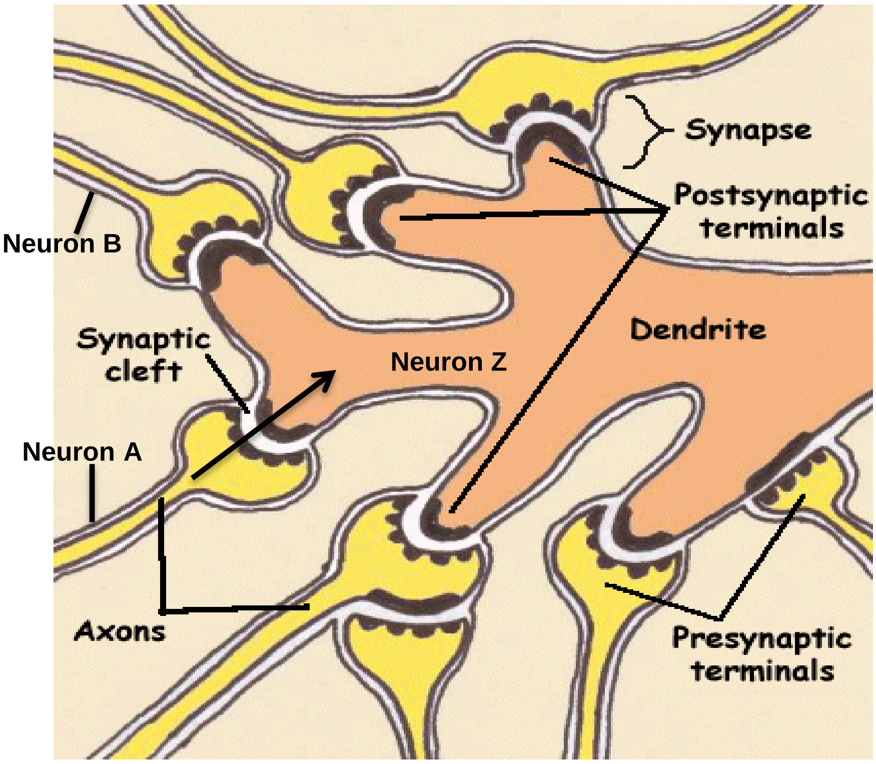

Fig. 8.1

(a) With reference to Fig. 8.1, describe how transmission of nerve impulse across the synapse is believed to have taken place.

Arrival of nerve impulse at synaptic knob depolarises presynaptic membrane of neuronA and increases permeability to Ca2+, causing influx of Ca2+;
Synaptic vesicles move, fuse with presynaptic membrane to discharge neurotransmitters by exocytosis;
Neurotransmitter diffuses across the synaptic cleft to attach to specific receptor sites on post-synaptic membrane; Ligand gated ion channels opens up to allow influx of Na+;
This bring about local depolarisation which when exceeds threshold, an action potential is triggered in the axon hillock of post-synaptic neuron Z;

; for 1 mark, max is 3 marks

For Examiner's Use

When brain experiences an abundance of stress, it can be caused by a surplus of adrenaline. To neutralize this extra adrenaline, the brain produces neurotransmitters, one of which is GABA, that have inhibitory effects upon the nervous system.

Neurone **A** releases adrenaline, a neurotransmitter which stimulates the production of nerve impulses in neurone **Z**. Neurone **B** releases GABA, which inhibits the production of nerve impulses in Neurone **Z**.

(b) Explain how Neurone **B** inhibits the production of nerve impulses in neurone **Z**.

Release of GABA into synaptic cleft binds & opens ligand-gated K+ or Cl- channels on post-synaptic neuronal membrane Z;
leading to K+ efflux OR Cl- influx for the postsynaptic neuron Z;
leading to hyperpolarisation < -70mV;
generates IPSP;
; for 1 mark, max is 2 marks

[ ]

(c) Describe one mechanism that helps to restore the resting membrane potential

Active transport involving Na^+- K^+ pump on membrane ;
$3Na^+$ out and $2K^+$ in + net loss of positive charges from the neurone ;
Establishes concentration gradient across neurone membrane ;

- Inside → high $[K^+]$, low $[Na^+]$
- Outside → low $[K^+]$, high $[Na^+]$

Facilitated diffusion involving Na^+ and K^+ ungated ion channels ;
More ungated K^+ channels on membrane + more K^+ diffuses out of neurone (than Na^+ diffuse in) ;
Net loss of positive charges from the neurone ;

; for 1 mark, max is 2 marks

[ ]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

9.

(a) Describe the structure of phospholipids and explain how the structure enables them to form the cell surface membrane. [4]

Structure:
It is made up of one glycerol, two fatty acids and one phosphate group (PO_4^{3-});
The fatty acids are joined to the glycerol group via ester bonds, while the phosphate group is joined to glycerol via phosphoester bond;
Since it has both hydrophobic and hydrophilic regions, it is also known as an amphipathic molecule;

; for 1 mark, max is 2 marks

Formation of cell surface membrane:
Hydrophilic phosphate heads interact with the aqueous medium/environment;
Hydrophobic interactions between fatty acid chains/tails;
in an aqueous environment, phospholipids self-assemble into a bilayer, forming a hydrophobic boundary/barrier that separates the living cells from its surrounding;

; for 1 mark, max is 2 marks

(b) Describe the role of nicotinamide adenine dinucleotide in aerobic and anaerobic respiration. [7]

Aerobic Respiration

NAD^+ is a coenzyme involved in redox reactions and functions as a mobile electron (and proton) carrier to carry high energy electrons and H^+ from organic molecules to the electron transport chain on the cristae of mitochondria;

Organic molecules are oxidized during glycolysis, link reaction and Krebs cycle and the electrons and H^+ from the oxidation process are transferred to NAD^+ to form NADH;

The electrons in NADH are used to reduce the electron acceptors of the electron transport chain, while NADH itself gets re-oxidised;

As electrons pass down the chain, the release of energy in a series of redox reactions is coupled to the phosphorylation of ADP to form ATP;

The protons liberated in the oxidation of NADH are used to establish the proton motive force necessary for ATP synthesis;

Reoxidation of 1 NADH yields 3 ATP via oxidative phosphorylation;

Reoxidation of NADH allows the regeneration of the coenzyme NAD^+, allowing it to pick up more protons and electrons from the Krebs cycle, link reaction and glycolysis so that these reactions can continue;

; for 1 mark, max is 6 marks
Anaerobic Respiration

NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;

NAD reused in glycolysis to generate 2 ATP by substrate level phosphorylation;
; for 1 mark, max is 7 mark

For Examiner's Use

(c) Describe the structure of the deoxyribonucleic acid and explain how they are suited for replication. [9]

DNA molecule consists of two polynucleotide chains that coils around each other to form a double helix;
Each chain consists of a sugar-phosphate backbone which is held together by phosphodiester bonds;
Made up of 4 types of deoxyribonucleotides: A, G, C, T;
Two polynucleotide chains are held together by weak hydrogen bonds between their nitrogenous bases;
Regular width 2nm of the double helix due to pairing between purine and pyrimidine bases;
2 hydrogen bond for every A-T base pairing and 3 hydrogen bonds for every C-G base pairing;
Two polynucleotide chains are anti-parallel;

; for 1 mark, max is 5 marks

Phosphodiester bonds, H bonds and hydrophobic interactions between stacked bases confer stability to the structure/integrity of the structure;
DNA replication requires one polynucleotide chain to act as a template strand to synthesise a daughter strand by complementary base-pairing/Allows for semi conservative replication;
Less energy is needed to overcome the 'weak' hydrogen bonds easily;
DNA molecule can unwind and separate by helicase;
Replication starts at the origin of replication, which is A-T rich;
Replication forks form at both directions from the origin of replication, resulting in bidirectional replication;
Antiparallel nature result in leading and lagging strands synthesis as DNA polymerase can only add nucleotides to the free 3'OH end of the DNA strand;

[Total: 20]

10.
(a) With reference to the key stages of cell signalling, describe how a liver cell respond to insulin stimulation. [7]

	Stages	Insulin
1.	Reception (max is 2 marks)	The binding of insulin to receptor tyrosine kinase causes two receptor monomers/subunits to dimerise;
2.		This activates the tyrosine kinase region of each polypeptide, which uses ATP to cross-phosphorylate the tyrosine residues on each other's cytoplasmic tails;
3.	Transduction (max is 2 marks)	Activation of relay proteins;
4.		Generation of 2nd messengers e.g. release of calcium ions for sarcoplasmic reticulum
5.	Response (max is 3 marks)	Increase permeability of the liver cell to glucose by increasing the number of GLUT4 transporters on the cell surface.
		Increases / activates enzymes involved in formation of glycogen from glucose / glycogenesis;
6.		Inhibits glycogenolysis and gluconeogenesis;
7.		Increases uptake of amino acids for the synthesis of amino acids / uptake of fatty acids for the synthesis lipids;
8.		Increases rate of respiration;
9.		Increases rate of DNA and RNA synthesis;

; for 1 mark, max is 7 marks

For Examiner's Use

(b) Outline the main stages of Calvin cycle. [8]

2Calvin cycle occurs in the stroma of chloroplasts;

During CO_2 fixation, ribulose bisphosphate (RuBP) accepts CO_2 to form an unstable 6C intermediate;

catalyzed by ribulose bisphosphate carboxylase-oxygenase (Rubisco);

Unstable 6C intermediate immediately breaks down to 2 molecules of 3C compound known as glycerate phosphate / phosphoglycerate (GP / PGA);

GP can be converted to pyruvate which is used to synthesize fatty acids/amino acids;

During reduction, GP is then converted to triose phosphate / glyceraldehyde phosphate (TP / GALP) using energy from hydrolysis of ATP and reducing power from reduced NADP, produced by light dependent reactions;

TP contains more chemical energy than PGA / GP, and is the first carbohydrate made in photosynthesis to leave the Calvin cycle;

5/6 of the total amount of TP is used to regenerate the RuBP consumed in the first

reaction which requires energy from the hydrolysis of ATP;

1/6 of the total amount of TP is used to make other carbohydrates and glycerol;

2 rounds of the cycle is required to produce 1 molecule of glucose;

; for 1 mark, 8 marks max

(c) Explain the relationship between the frequency of sickle cell anaemia allele and the number of malaria cases. [5]

Frequency of sickle cell anaemia allele is highest in regions of endemic malaria;
Individuals with two recessive Hb^S alleles/Hb^SHb^S develop sickle cell disease as sickle red blood cells exhibit poor oxygen transport;
Homozygous individuals with two normal Hb^A alleles/Hb^AHb^A have normal red blood cells;
but have a greater risk of dying from malaria in regions of endemic malaria;
Heterozygous individuals with one recessive Hb^S allele/Hb^AHb^S produce both normal and abnormal haemoglobin;
Heterozygous condition hide recessive Hb^S allele that is less favourable from natural selection which only acts on sickle cell anaemia phenotypes;
thus maintaining the recessive allele in the population;
Heterozygote advantage: Heterozygotes are less likely to suffer from malaria thus they have selective advantage over homozygotes/heterozygote advantage;
Sickle shape of red blood cell results in lower oxygen carrying capacity thus malaria parasite cannot survive;
Or idea that malaria parasite cannot survive well in cell with Hb^S
Heterozygotes are more likely to survive and reproduce to pass on the Hb^S allele;

; for 1 mark, max is 5 marks

20]

End of Paper